

Multi-Catfish MCRL V0.25: Successor Decision Path

World Tape • Two Per-User Heads • Set-Level Coordinator over a Bounded Catalogue • Atomic Commit • 48-Boundary Endpoint

PRIMARY: a-r0

ENDPOINT: 48 BOUNDARIES

PRE-SEALED

I_heads — per user i, per legal action

i's candidate nominal geometry • i's history • previous served set excluding i (b^-_{-i}) • no joint physics • one forward pass

1. World Tape and Per-User Rows

World Tape (built once per world)

33 steps $\times$ 48 boundaries at $t + k(0.640 \text{ s})$
Geometry, channel and cross gains precomputed
Reused by every arm and every setting

Per-User Candidate Rows

28 association slots (4 satellites $\times$ 7 cells)
NO_OP legal exactly for an empty mask
One row per user per legal action

Frozen Row Schemas

Q1: rate-target margin, required power / cap, mode SE, occupancy excluding focal, angle, times
Q2: 22 fields; background b^-_{-i} at decision time
Nominal model only — realised fading is excluded

2. Two Heads → Per-User Proposal

Q_1 : Focal Opening Head

Pairwise zero-bootstrap regression
Supervised surrogate, not a Bellman value

Q_2 : Continuation / Forecast Head

Same training form, 22-field schema
Certificate is forecast validity

Per-User Masked Argmax

argmax over $\mathcal{A}_u^+$ of normalised $Q_1 + Q_2$
One forward pass per candidate row

a^0 (BASE) — validated profile

Independent picks repaired into a legal profile
Built BEFORE the coordinator's clock starts

I_coordinator — per anchor

global nominal geometry + beam cross gains
all legal sets • b^- • a^0 • $\mathcal{C}$ • $\mathcal{M}$

3. Set-Level Layer over the Catalogue

Bounded Catalogue $\mathcal{C}$ ($\approx 1\,200$ rows)

≤ 800 unilateral: top-8 options per user
 ≤ 180 pairwise: top-10 users $\times$ top-2 options
 ≈ 40 per-active-beam evacuation sets
 ≤ 200 S0 proposals and evacuations

Set-Level Selectors

S3 learned $\Psi_\theta(Z, a^0, A, a_A)$, deployed
permutation-invariant, $\Psi(\emptyset) = \Psi(\{u\}) = 0$
S0 the same selector with exact Ψ_A
S_UNI iterated exact unilateral comparator

Two-Stage Scoring

Stage 1 $C_1 + \Psi$ over all rows at $k = 0$
Stage 2 C_2 continuation over the top $M = 64$
Select the argmax of the complete score

10 s • 4 WORKERS

selected profile

4. Validation, Execution and the 48-Boundary Endpoint

Validation

Joint legality and repair
Service guard: no served-count decrease versus BASE

Coupled Capped Solve

$p \leftarrow \min(p^+, \Gamma(N_0 W + I(p))/\hat{h})$
Unique capped fixed point;
CONVERGED / CONVERGED_SLOW

Atomic Commit

One complete profile committed
Never split into per-user adoptions

48-Boundary Endpoint

〈結果待填〉

deadline miss → commit the pre-validated a^0 (counted in B and E)

NOMINAL → REALISED

Reading the boundaries: